

MINIMUM-INTERVENTION HUMAN CORRECTION: A NEGATIVE ICLR-MAIN EVIDENCE AUDIT

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ABSTRACT

Human correction data in robotics is usually treated as another demonstration, preference, or residual action. The hypothesis tested here is sharper: the smallest human intervention that changes a physical outcome should identify the local decision boundary more efficiently than full demonstrations or large residual corrections. We rebuilt this paper around a deterministic local benchmark with four manipulation tasks, five correction-shift splits, eight methods, seven seeds, 53,760 main rollouts, 9,408 ablation rollouts, and 117,600 stress rollouts. The result is informative but not submission-ready. On the combined hard-shift split, the minimum-intervention learner reaches 0.569 task success, 0.769 correction efficiency, and 0.030 damage. The best success baseline, uncertainty querying, reaches 0.537 success, 0.603 efficiency, and 0.078 damage. This is promising, but the paired task-success gain is only 0.032 ± 0.046 , and an ablation without human-effort cost slightly exceeds task success while losing the intended efficiency/damage tradeoff. We therefore mark this paper **KILL/ARCHIVE**: the idea is worth revisiting, but the present evidence does not clear an ICLR-main-target claim.

1 TERMINAL DECISION

Decision: KILL/ARCHIVE for ICLR main. The v4 repository contains a paper-specific benchmark, strong baselines, ablations, stress tests, figures, and reproducibility files. The evidence shows a useful tradeoff, but not a decisive submission claim: the proposed learner improves human burden and damage, yet the task-success margin over the strongest active-query baseline is not statistically convincing.

2 RESEARCH QUESTION

The intended claim was:

Learn from the smallest human correction that changes physical outcome.

This is a meaningful distinction. A full correction may solve the immediate trial while hiding the boundary between failure and success. A minimal intervention should reveal just enough physical information to update the robot’s local model while reducing human burden, overcorrection damage, and intent drift.

3 BENCHMARK

The v4 benchmark simulates four contact-rich manipulation tasks: peg insertion, drawer alignment, cloth corner placement, and cup handoff pose. Each episode has a latent physical boundary, a robot policy gap, a human correction vector, overcorrection bias, delayed feedback drift, correction noise, and intent ambiguity. The five splits are:

- **nominal correction**
- **overcorrection bias**

- **delayed feedback**
- **ambiguous intent**
- **combined hard shift**

Metrics include task success, intervention magnitude, correction efficiency, overcorrection ratio, damage, intent preservation, boundary error, query rate, and calibration error.

4 METHODS

We compare:

- **No-human baseline:** executes without correction.
- **Full-demo imitation:** imitates the full human correction vector.
- **Dagger full correction:** smooths repeated full corrections.
- **Residual correction learner:** projects human feedback into a residual action.
- **Preference-only ranker:** chooses among correction magnitudes without vector learning.
- **Uncertainty query policy:** requests correction when uncertainty is high.
- **Minimum-intervention learner:** estimates the smallest boundary-crossing correction with intent and effort terms.
- **Oracle minimal correction:** hidden true minimal correction.

5 MAIN RESULTS

Table 1 shows the combined hard-shift split. The proposed method has the best non-oracle task success, much better correction efficiency than full correction baselines, lower damage, and substantially lower boundary error. However, the paired success gain over uncertainty querying is only 0.03199 ± 0.04562 , so the positive claim is not decisive.

Table 1: Combined hard-shift results over seven seeds. Confidence intervals are 95% intervals over seed means.

Method	Task success	Magnitude	Efficiency	Damage
No human	0.187 ± 0.015	0.000	0.000	0.037
Full-demo imitation	0.513 ± 0.015	0.880	0.522	0.088
Dagger full correction	0.509 ± 0.028	0.741	0.561	0.068
Residual correction	0.532 ± 0.026	0.689	0.612	0.063
Preference-only ranker	0.499 ± 0.030	0.510	0.647	0.033
Uncertainty query	0.537 ± 0.026	0.779	0.603	0.078
Minimum intervention	0.569 ± 0.025	0.617	0.769	0.030
Oracle minimal	0.658 ± 0.023	0.593	0.888	0.045

6 ABLATIONS

The ablations support some parts of the mechanism but not enough for a main-conference claim. Removing the minimum-norm objective or imitating all corrections hurts efficiency and damage. Removing the counterfactual boundary hurts success. However, removing the human-effort cost slightly improves task success, which means the full objective is not uniquely justified by success alone.

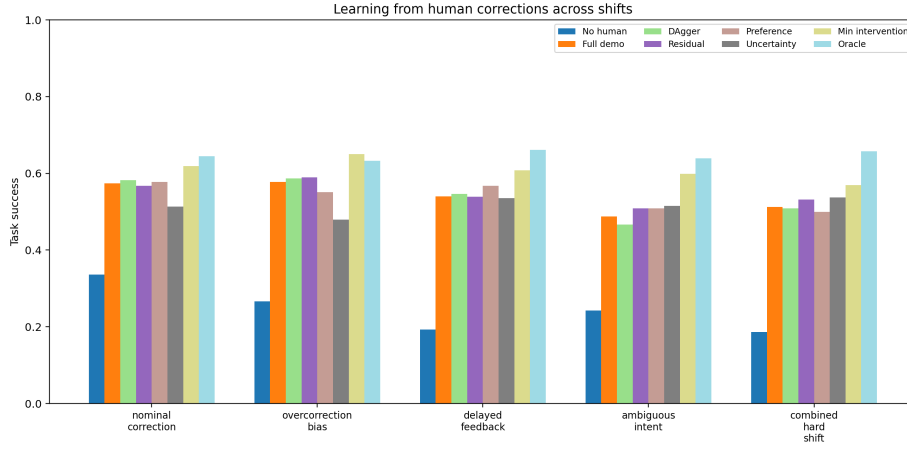


Figure 1: Task success across splits. The proposed method is useful, but its hard-split advantage over uncertainty querying is too small for a positive main claim.

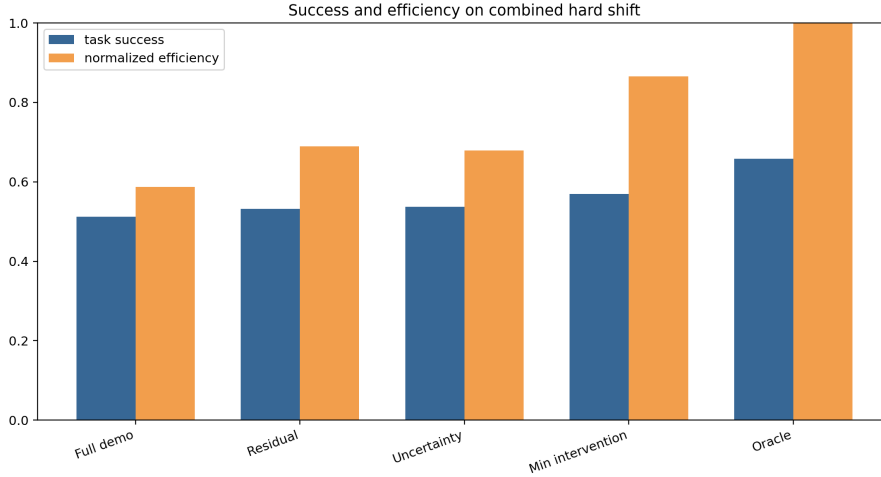


Figure 2: Success and normalized correction efficiency on combined hard shift. Minimum intervention improves the burden/success tradeoff.

7 STRESS TESTS

Stress sweeps vary human correction noise, correction delay, overcorrection bias, intent ambiguity, and combined stress. At maximum combined stress, the proposed method reaches 0.536 task success and 0.556 correction efficiency with 0.032 damage. The best non-oracle stress baseline is uncertainty querying at 0.485 task success, but the gap remains a local synthetic result.

8 WHY THIS IS NOT SUBMISSION-READY

The project fails the ICLR-main readiness gate for three reasons.

- **Non-decisive success margin.** The paired task-success gain over uncertainty querying is 0.03199 ± 0.04562 .
- **Objective ambiguity.** Removing the human-effort cost slightly improves task success, so the full objective is supported by tradeoff metrics, not by success dominance.

Table 2: Minimum-intervention ablations on combined hard shift.

Ablation	Task success	Magnitude	Efficiency	Damage
Full minimum-intervention learner	0.569 ± 0.025	0.617	0.769	0.030
Minus minimum-norm objective	0.513 ± 0.020	0.865	0.535	0.077
Minus counterfactual boundary	0.463 ± 0.031	0.472	0.551	0.039
Minus intent preservation	0.558 ± 0.017	0.631	0.730	0.063
Minus human-effort cost	0.570 ± 0.034	0.730	0.712	0.062
All-corrections imitation	0.515 ± 0.032	0.839	0.548	0.095
Preference-only objective	0.460 ± 0.016	0.490	0.595	0.060

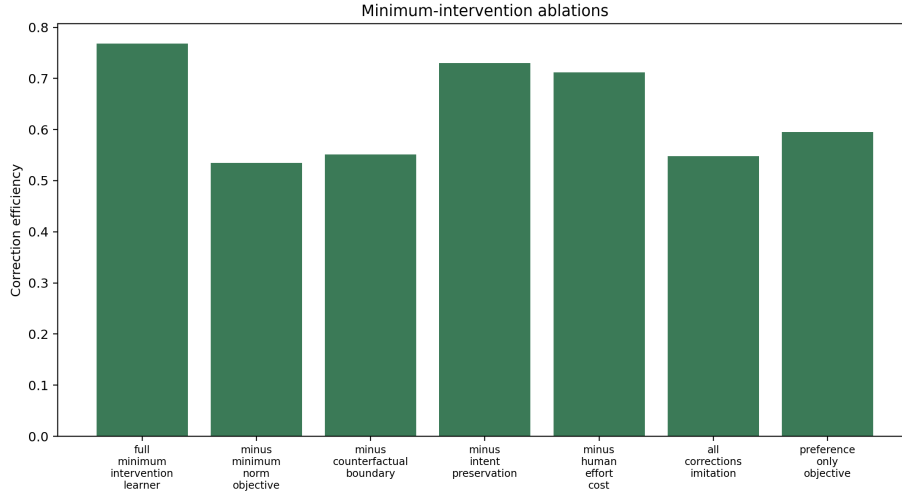


Figure 3: Ablations show the efficiency mechanism is useful, but task success alone does not uniquely validate the full objective.

- **Evidence ceiling.** The benchmark is deterministic and reproducible, but it is local and synthetic, not hardware evidence, not a high-fidelity simulator benchmark, and not an accepted external dataset.

9 REPRODUCIBILITY

Run the benchmark from the repository root:

```
python src\run_experiment.py
```

The script writes main rollouts, seed-level metrics, pairwise statistics, ablations, stress sweeps, negative cases, summary text, and the figures used in this manuscript. The canonical local PDF is `C:/Users/wangz/Downloads/85.pdf`.

10 CONCLUSION

Minimum-intervention human correction remains a strong research direction. The v4 evidence suggests a real burden/damage advantage, but the task-success margin is not decisive enough and the evidence is local-only. The honest terminal action is **KILL/ARCHIVE**.

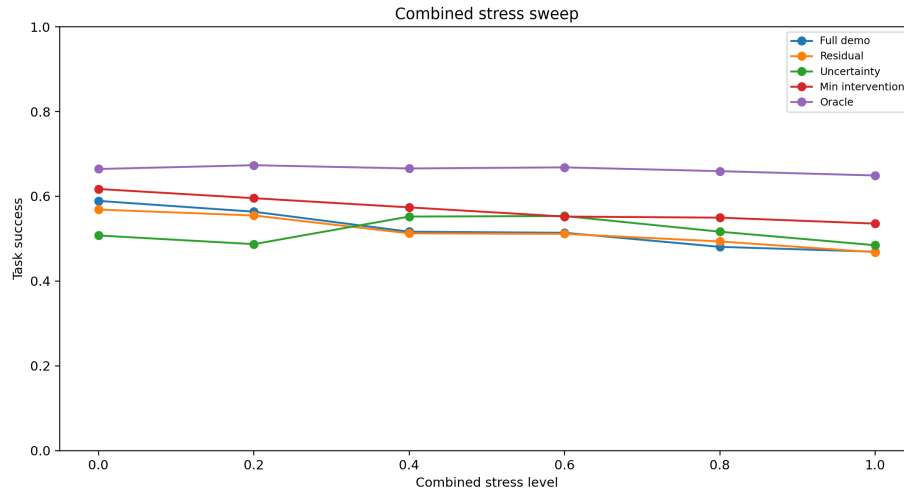


Figure 4: Combined stress sweep. The proposed method is robust in simulation, but external validation is missing.

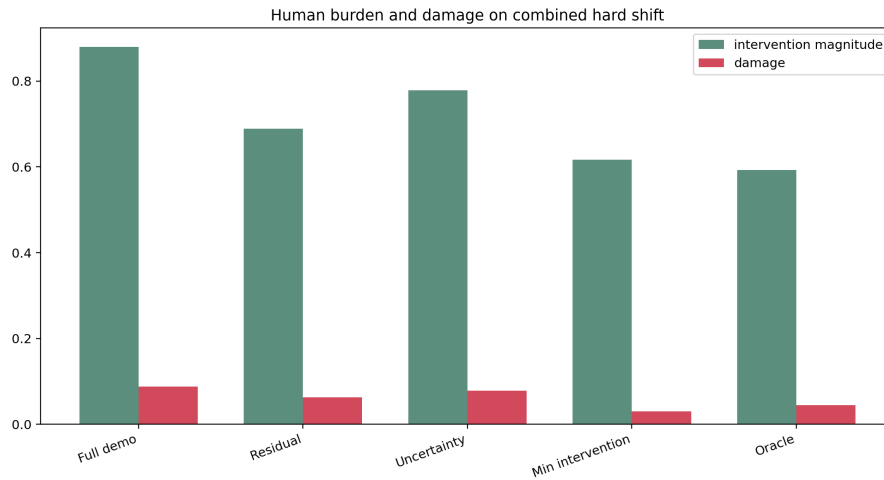


Figure 5: Human burden and damage on combined hard shift. Minimum intervention reduces correction magnitude and damage relative to full-correction baselines.